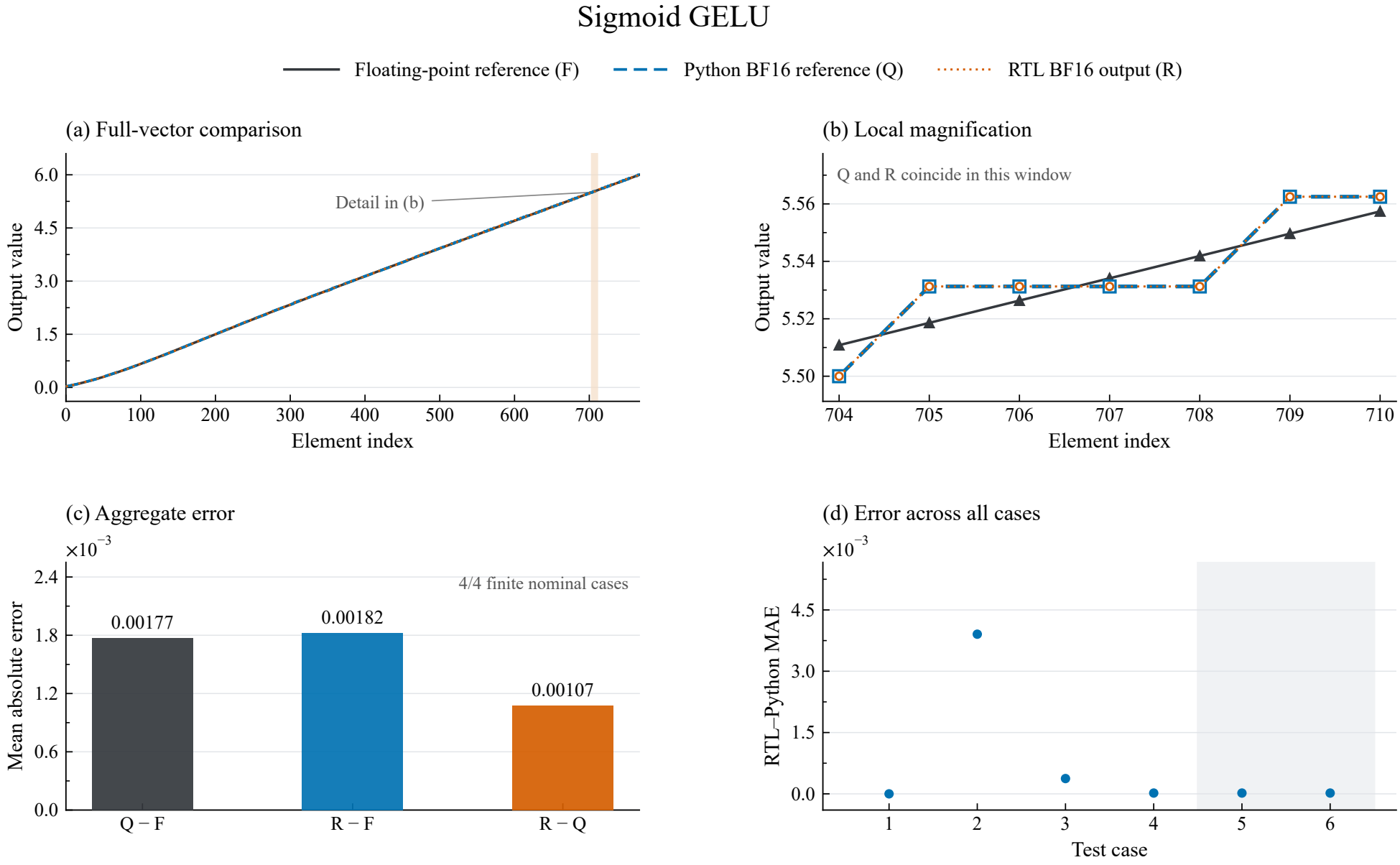
Sigmoid GELU
Floating-point reference (F)
Python BF16 reference (Q)
RTL BF16 output (R)
(a) Full-vector comparison
Output value
Element index
Detail in (b)
(b) Local magnification
Q and R coincide in this window
Output value
Element index
(c) Aggregate error
×10⁻³
Mean absolute error
4/4 finite nominal cases
0.00177
0.00182
0.00107
Q − F
R − F
R − Q
(d) Error across all cases
×10⁻³
RTL–Python MAE
Test case
GEMM + operator · N = 768 · ramp example #3 · shaded cases: probes · red markers: failed checks
All nominal cases are finite; zero RTL–Python error means numerical equality for those samples.